SmartMove Live Viva Practical Demonstration & Training Manual

Step-by-Step Viva Execution Guide, Database Demonstrations, Speaking Scripts & Rubric Defense

Member & ID	Student Name	Assigned Viva Scope & Software Tools	Key Responsibilities
Member 1 (037)	J.H.N. Fernando	Oracle 21c XE / SQL Developer / cmd	Master Relational Schema, 3NF DDL, Constraints, Sequences, Roles, Da
Member 2 (036)	Arshath MLM	Oracle XE / PL/SQL / mongosh	CALCULATE_FARE Function, Materialized Views, Reports 1-3, MongoDB
Member 3 (028)	D.D. Kandeepan	Oracle XE / PL/SQL Engine	TRG_MANAGE_SEATS Trigger, Concurrency Locking (SELECT FOR UP
Member 4 (004)	K.G.L. Sandanimne	MongoDB Compass / mongosh / React UI	MongoDB Collections, JSON Validators, Query 3 (Text Search), Query 4 (

1. Assessment Structure & Rubric Breakdown (from DM2 CW -1 (1).pdf)

According to the official coursework document, the evaluation consists of **Software Application Implementation (80%)** and **Presentation & Live Viva (20%)**. Examiners evaluate working software and live database query execution:

Rubric Criteria	Weight	70%+ Top Distinction Band Expectation (What You Must Demonstrate)
Front-end Development + ER Diagram	20%	Fully functional user-friendly frontend covering all 9 operations; complete normalized 3NF ER diagram.
Database Implementation (Oracle)	15%	All operations implemented with PL/SQL, correct schema, strong RBAC security and backup mechanism.
MongoDB Incorporation	15%	Correct MongoDB incorporation, schema-flexible structure, and the 4 standard query scenarios fully captured.
Reports & Business Logic (PL/SQL)	15%	Comprehensive reports (at least 5), robust PL/SQL logic, and advanced exception handling.
Completeness of the Project	5%	Fully complete, all requirements achieved, no missing operational modules.
Integration & Innovation	10%	Strong integration between Oracle and MongoDB for hybrid analytics and real-time content delivery.
Presentation & Viva	20%	Excellent individual understanding, confident technical communication, demonstrates deep knowledge.

2. Pre-Flight Setup Checklist (10 Minutes Prior to Viva)

Open terminal in project root and execute the silent start script: ./start_full_system.ps1. Ensure 4 windows are open:

- **Window 1 (Oracle SQL Developer / SQL*Plus):** Connected to service XEPDB1 as user SMARTMOVE.
- **Window 2 (MongoDB Shell / Compass):** Connected to mongodb://localhost:27017/smartmove.
- **Window 3 (Web Browser):** Tab 1 on http://localhost:3000; Tab 2 on http://localhost:8761 (Eureka Registry).
- **Window 4 (VS Code):** Open to database_SQL_noSQL_PLSQL/ showing clean student scripts.

3. Deep-Dive: The 4 Mandatory MongoDB Queries (Page 3 of CW Brief)

These 4 exact queries are explicitly mandated in the assessment specification. Practice running them in mongosh:

Query 1: Retrieve All Passenger Reviews for a Specific Route

Scenario: Operational managers inspect star ratings, comments, and feedback tags for Route 1 (Colombo to Kandy).

```
db.reviews.find(
    { routeId: 1 },
    { _id: 0, passengerId: 1, rating: 1, type: 1, tags: 1, comment: 1, createdAt: 1 }
).sort({ createdAt: -1 });
```

Speaking Script: *"Query 1 retrieves customer feedback for Route 1 sorted by date. In relational databases, storing tags and optional fields requires multi-table junction schemas. In MongoDB, documents naturally store dynamic arrays like ['On Time', 'Clean']. We created an index on routeId to guarantee O(log N) lookup time."*

Query 2: Identify Highest-Rated Vehicles or Drivers Based on Customer Feedback

Scenario: Fleet management computes mean rating and review volume to reward top drivers and buses.

```
db.reviews.aggregate([
    { $match: { vehicleId: { $ne: null } } },
    { $group: { _id: "$vehicleId", averageRating: { $avg: "$rating" }, totalReviews: { $sum: 1 } } },
    { $project: { _id: 0, vehicleId: "$_id", averageRating: { $round: ["$averageRating", 2] }, totalReviews: 1 } },
    { $sort: { averageRating: -1, totalReviews: -1 } },
    { $limit: 5 }
]);
```

Speaking Script: *"Query 2 demonstrates a 5-stage aggregation pipeline ($match -> $group -> $project -> $sort -> $limit). It calculates arithmetic mean ($avg) and review volume ($sum), rounds ratings using $round, and limits output to the top 5 vehicles. To rank drivers instead, we simply group by $driverId."*

Query 3: Search Customer Complaints or Feedback Using Specific Keywords

Scenario: Customer service managers search thousands of reviews for critical keywords ('delay', 'breakdown') using text indexes.

```
// Prerequisite: Text index on comment field: db.reviews.createIndex({ comment: "text" });
db.reviews.find(
    { $text: { $search: "delay breakdown" } },
    { score: { $meta: "textScore" }, _id: 0, passengerId: 1, routeId: 1, rating: 1, type: 1, comment: 1 }
).sort({ score: { $meta: "textScore" } });
```

Speaking Script: *"Query 3 utilizes MongoDB's Full-Text Search. Instead of slow regex scans (LIKE %delay%), it uses an inverted text index on comment. It computes a relevance score using {$meta: 'textScore'} and sorts descending by score, allowing support staff to triage urgent breakdowns instantly."*

Query 4: Retrieve Vehicle Documents and Multimedia Information

Scenario: Fleet inspectors check vehicle compliance certificates; dispatch officers inspect trip departure photos and dashcam video.

```
// 4A: Retrieve legal certificates and photos for Vehicle 1 (Embedded Document Pattern)
db.vehicle_documents.find({ vehicleId: 1 }, { _id: 0, vehicleId: 1, registration: 1, documents: 1, images: 1 });

// 4B: Retrieve operational multimedia for Trip 1
db.trip_media.find({ tripId: 1 }, { _id: 0, tripId: 1, routeId: 1, mediaType: 1, caption: 1, url: 1 });
```

Speaking Script: *"Query 4 demonstrates the Embedded Document Pattern. In Oracle, vehicle certificates require 3 tables and complex SQL JOINs. In MongoDB, insurance, revenue licenses, and images are embedded in a single parent envelope, fetched in an atomic single-read operation. Media files are stored in cloud storage (S3/CDN) with metadata in MongoDB."*

4. Individual Member Live Viva Practice Drills

Each team member must deliver a confident 20-second opening statement and execute their live tasks:

Member 1 (037 — J.H.N. Fernando): Master Schema, Constraints, Security & Backup

Opening Pitch: *"I designed the 4 master foundation tables in 3NF, implemented sequence-based surrogate keys to eliminate lock contention, enforced database-level check constraints, configured smartmove_app_role for least-privilege security, and formulated our Data Pump backup plan."*

```
-- Show Master Constraints:
SELECT TABLE_NAME, CONSTRAINT_NAME, CONSTRAINT_TYPE FROM USER_CONSTRAINTS WHERE TABLE_NAME IN ('PASSENGER','DRIVER','ROUTE','
VEHICLE');
-- Show Sequences:
SELECT SEQUENCE_NAME, INCREMENT_BY, LAST_NUMBER FROM USER_SEQUENCES;
-- Show Security Role Privileges:
SELECT * FROM ROLE_TAB_PRIVS WHERE ROLE = 'SMARTMOVE_APP_ROLE';
-- Data Pump Export command:
expdp system/oracle@localhost:1521/XEPDB1 schemas=SMARTMOVE directory=DATA_PUMP_DIR dumpfile=smartmove_full.dmp logfile=smart
move_exp.log
```

Member 2 (036 — Arshath MLM): PL/SQL Function, Materialized Views & Reports 1–3

Opening Pitch: *"I created the CALCULATE_FARE function with exception handling, built MV_MONTHLY_REVENUE Materialized View for instant monthly revenue aggregation, developed procedure REPORT_REVENUE, and built Reports 1 to 3 using window ranking functions."*

```
-- 1. Execute CALCULATE_FARE:
SELECT CALCULATE_FARE(1, 2) AS TOTAL_FARE FROM DUAL; -- LKR 3000.00
-- 2. Show Exception Handling:
SELECT CALCULATE_FARE(999, 1) FROM DUAL; -- Raises ORA-20001: Route ID 999 does not exist.
-- 3. Query Materialized View:
SELECT * FROM MV_MONTHLY_REVENUE;
-- 4. Report 1 (Window Ranking):
SELECT r.ROUTE_ID, r.ORIGIN, r.DESTINATION, COUNT(b.BOOKING_ID) AS BOOKINGS,
       RANK() OVER (ORDER BY COUNT(b.BOOKING_ID) DESC) AS ROUTE_RANK
FROM ROUTE r LEFT JOIN TRIP t ON r.ROUTE_ID = t.ROUTE_ID
LEFT JOIN BOOKING b ON t.TRIP_ID = b.TRIP_ID AND b.STATUS != 'CANCELLED'
GROUP BY r.ROUTE_ID, r.ORIGIN, r.DESTINATION ORDER BY ROUTE_RANK;
```

Member 3 (028 — D.D. Kandeepan): Transactional Engine, Trigger & Concurrency

Opening Pitch: *"I developed the transactional tables—Trip, Booking, Payment, Maintenance, Feedback. I implemented TRG_MANAGE_SEATS trigger to maintain seats automatically, implemented MAKE_BOOKING with SELECT ... FOR UPDATE pessimistic locking, and built Reports 4 and 5."*

```
-- THE GOLD TRIGGER DEMO (Run live in SQL Developer):
-- Step 1: Check seats on Trip 1 (Shows 43)
SELECT TRIP_ID, AVAILABLE_SEATS FROM TRIP WHERE TRIP_ID = 1;
-- Step 2: Book 2 seats
INSERT INTO BOOKING (BOOKING_ID, PASSENGER_ID, TRIP_ID, SEATS_BOOKED, TOTAL_FARE, STATUS, PAYMENT_STATUS)
VALUES (SEQ_BOOKING.NEXTVAL, 1, 1, 2, 3000.00, 'CONFIRMED', 'UNPAID'); COMMIT;
-- Step 3: Check seats again (Automatically decreased to 41!)
SELECT TRIP_ID, AVAILABLE_SEATS FROM TRIP WHERE TRIP_ID = 1;
-- Step 4: Cancel booking
UPDATE BOOKING SET STATUS = 'CANCELLED' WHERE BOOKING_ID = (SELECT MAX(BOOKING_ID) FROM BOOKING); COMMIT;
-- Step 5: Check seats again (Automatically restored to 43!)
SELECT TRIP_ID, AVAILABLE_SEATS FROM TRIP WHERE TRIP_ID = 1;
```

Member 4 (004 — K.G.L. Sandanimne): MongoDB Architecture, Full-Text & Web App

Opening Pitch: *"I architected the MongoDB document database layer with JSON Schema validators, implemented full-text indexing for Query 3 and nested media retrieval for Query 4, and connected the React frontend to our 12 Spring Boot microservices for real-time operations."*

```
-- Live Web Application Flow (http://localhost:3000):
1. Login as Passenger (nimal@example.com / password123) -> Search trips -> Book 2 seats.
2. Open Incognito window, login as Driver (saman.k@smartmove.lk) -> Trip appears in real-time -> Click 'Start Journey'.
3. In Passenger tab -> Click 'Pay with Card' -> Status becomes COMPLETED.
4. Login as Admin (admin@smartmove.lk) -> Open /admin/reports -> Click 'Export PDF' to show live business reports!
```

5. Slide-by-Slide Presentation Allocation (15 Minutes)

Slide Range	Speaker	Slide Topics & Content	Viva Focus
Slides 1–4	Member 1 (037)	Introduction, Problem Statement, Objectives, Core System Operations	Business Context
Slides 5–7	Member 1 (037)	Conceptual EER Diagram, Relational Schema (3NF Normalization), Constraints	Data Modeling & Integrity
Slides 8–10	Member 2 (036)	PL/SQL Fare Calculation Engine, Materialized Views, Executive Reports 1–3	Database Computing
Slides 11–13	Member 3 (028)	Transactional Architecture, TRG_MANAGE_SEATS Trigger, Concurrency Locking	ACID & State Machine
Slides 14–16	Member 4 (004)	MongoDB Polyglot Persistence, Collections, Full-Text Search, Media Retrieval	NoSQL Document Modeling
Slides 17–19	Member 4 (004) / All	Full-Stack System Architecture, Live Web Demonstration, Conclusion & Q&A	System Integration

6. Top 10 Viva Curveball Questions & Model Answers

#	Examiner Question	High-Scoring Model Answer
1	Why use both Oracle and MongoDB?	Polyglot Persistence: Oracle guarantees ACID consistency for financial transactions and seat availability. MongoDB provide
2	Why calculate fares in PL/SQL rather than frontend?	Calculating fares inside the database ensures business rule integrity. Client-side code can be inspected or bypassed by AP
3	How do you prevent overbooking when two users book simultaneously?	In MAKE_BOOKING, we issue SELECT ... FOR UPDATE on the TRIP row. This acquires an exclusive row-level lock. The
4	What is the difference between View and Materialized View?	A standard view is a saved virtual query executed every time it is queried. A Materialized View computes the aggregate befor
5	What is the difference between Statement and Row-level triggers?	Statement triggers fire once per SQL statement. Row-level triggers (FOR EACH ROW) fire once for every affected row, pro
6	What is the purpose of NOCACHE on sequences?	By default, Oracle caches 20 sequence numbers in SGA memory. If the server restarts unexpectedly, cached numbers are
7	Can you rollback a CREATE TABLE or TRUNCATE?	No. In Oracle, DDL statements issue an implicit COMMIT before and after execution. Only DML statements (INSERT, UPD
8	How does MongoDB validate data without foreign keys?	We enforce collection contracts via $jsonSchema validators (data types, mandatory fields, enum constraints). Cross-databa
9	What is the difference between %TYPE and %ROWTYPE?	%TYPE anchors a variable to a single column's data type. %ROWTYPE declares a composite record variable holding an e
10	What is the difference between WHERE and HAVING in Report 4?	WHERE filters rows before aggregation. HAVING filters the aggregate results after GROUP BY. In Report 4, we must calcu

7. Emergency Troubleshooting Playbook

- **If Oracle service is stopped:** Open Admin PowerShell and run: Start-Service OracleServiceXE; Start-Service OracleOraDB21Home1TNSListener
- **If MongoDB service is stopped:** Open Admin PowerShell and run: Start-Service MongoDB
- **If ports (8080/3000) are blocked:** Run: ./stop_all.ps1 to cleanly kill stale processes and release all ports.